A kinetic-impact deflection model

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a velocity change and explain the value of lead time.

SUCCESS CRITERIA - I can...

1. State the simplified collision model.
2. Connect delivered momentum to target velocity change.
3. Explain why real deflection includes ejecta and orbital effects.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Two equal 3 kg carts approach at 5 m/s in opposite directions and stick. Find their common final velocity and explain.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Delta v \approx p_{projectile}/M_{target}$; shift ≈ Δv × time.
Toy model: much heavier target; straight-line shift.

Worked reasoning: In a fictional model, a 2 kg projectile at 30 m/s sticks to a stationary 10 kg target. Final speed is 60/12=5 m/s. The system includes both objects.

Your turn: Draw the before-and-after system boundary and momentum arrows.

Marzano check - circle: ① ② ③

03 Look and explain

A tiny sideways velocity change acts for a short time or a long time. Which produces more sideways shift in the straight-line model? Explain.

Marzano check - circle: ① ② ③

04 Build the solution

A supplied target velocity change is .01 m/s sideways. Calculate shift after 100 s and 1000 s.

Marzano check - circle: ① ② ③

05 Use the idea

Why should the no-ejecta sticking calculation not be reported as the exact measured result of a real mission?

Marzano check - circle: ① ② ③

06 Choose a method

An asteroid's period is 6 years. Explain why that number alone cannot establish an Earth encounter. Name two additional observations needed.

Marzano check - circle: ① ② ③

07 Exit check

At the same smallΔv, compare shift with twice as much lead time. State one model limit.

Marzano check - circle: ① ② ③

08 STRETCH: Apply it another way

Return to the worked example in problem 2. Keep projectile momentum 60 kg·m/s but double the stationary target mass from 10 to 20 kg. Calculate the new final speed and explain the change.

Marzano check - circle: ① ② ③